

# Timeline Check-in

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## Set up

### Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
library(ggribes)
```

## Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")
```

```
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪ Data-2026-03-10.xlsx"),
                          sheet = "Water Quality",
                          na = c("", "n/a", "N/A", "over", "173+"))
```

## Cleaning and Wrangling

### Cleaning

#### sites object

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

#### taxon\_list object

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake")) |>
  select(verbatim_name, scientificName)
```

#### water\_quality object

```
water_quality_clean <- water_quality |>

# cleaning column names
clean_names() |>

# filtering join: only include sites in ncos_sites
```

```

# keep all the rows on the left that match up with rows on the right
semi_join(ncos_sites, by = "site") |>

# combine date and site together and remove those indiv. columns
unite("date_site", date, site, remove = TRUE) |>

# combine date and site together and remove those indiv. columns
group_by(date_site) |>

# create new columns that show median of each variable at each date_site
# because there are multiple samples at the same site at different depths
summarize(
  med_ph = median(p_h, na.rm = TRUE),
  med_sal = median(salinity_ppt, na.rm = TRUE),
  med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE),
  med_temp = median(temperature_c, na.rm = TRUE)) |> # <-- ADD
  ↪ THIS

# ungroup
ungroup() |>

# take out outlier
filter(date_site != "2022-08-12_NVBR")

```

## aq\_ins object

```

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
  ostracod,
  copepod,
  cladocera) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format

```

```

pivot_longer(cols = ostracod:cladocera,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪ = TRUE),
       site_full = fct_rev(site_full)) |>

# ADD: log+1 transform of average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1)) |>                                     # <-- Fig
  ↪ 1, 2, 3, 4

# ADD: season derived from date
mutate(season = case_when(
  month(date) %in% c(3, 4, 5) ~ "Spring",
  month(date) %in% c(6, 7, 8) ~ "Summer",
  month(date) %in% c(9, 10, 11) ~ "Fall",
  month(date) %in% c(12, 1, 2) ~ "Winter"
),

```

```

season = factor(season, levels = c("Spring", "Summer", "Fall", "Winter"))
↪ |># <-- Fig 4
mutate(taxon = factor(taxon, levels = c("copepod", "ostracod",
↪ "cladocera")))

#ADD: capitalizing invertebrate names
aq_ins_clean$taxon <- str_to_sentence(aq_ins_clean$taxon)

```

## Setting plot aesthetics

```

# colors for taxa
taxon_cols <- c(
  "Ostracod" = "orange2",
  "Copepod" = "tomato3",
  "Cladocera" = "turquoise4")

sites_cols <- c(
  "NDC" = "hotpink3",
  "NEC" = "cyan3",
  "NMC" = "lightpink2",
  "NPB" = "slateblue3",
  "NPB1" = "steelblue3",
  "NPB2" = "green3",
  "NVBR" = "maroon3"

)

# setting common theme elements
theme_set(
  theme_classic() +
  theme(text = element_text(size = 12),
    plot.title.position = "plot",
    legend.position = "none"))

```

## Visualizations directly relevant to answering questions

Figure 1

```
# base layer: create new object to store figure 1
figure1 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
                                y = log_ave_lit,
                                color = taxon)) +

# first layer: points
geom_point() +

# facet by site
facet_wrap(~site_full) +

# labels
labs(title = "Figure 1:
Dominant Three Taxa Abundance and Average Water Temperature by Site",
      x = "Average Temperature (°C)",
      y = "log + 1 Transformed Average Abundance/L",
      color = "Taxon") +

# manual colors
scale_color_manual(values = taxon_cols) +

# adding legend
theme(legend.position = "right")

figure1
```

Figure 1:

Dominant Three Taxa Abundance and Average Water Temperature by Site

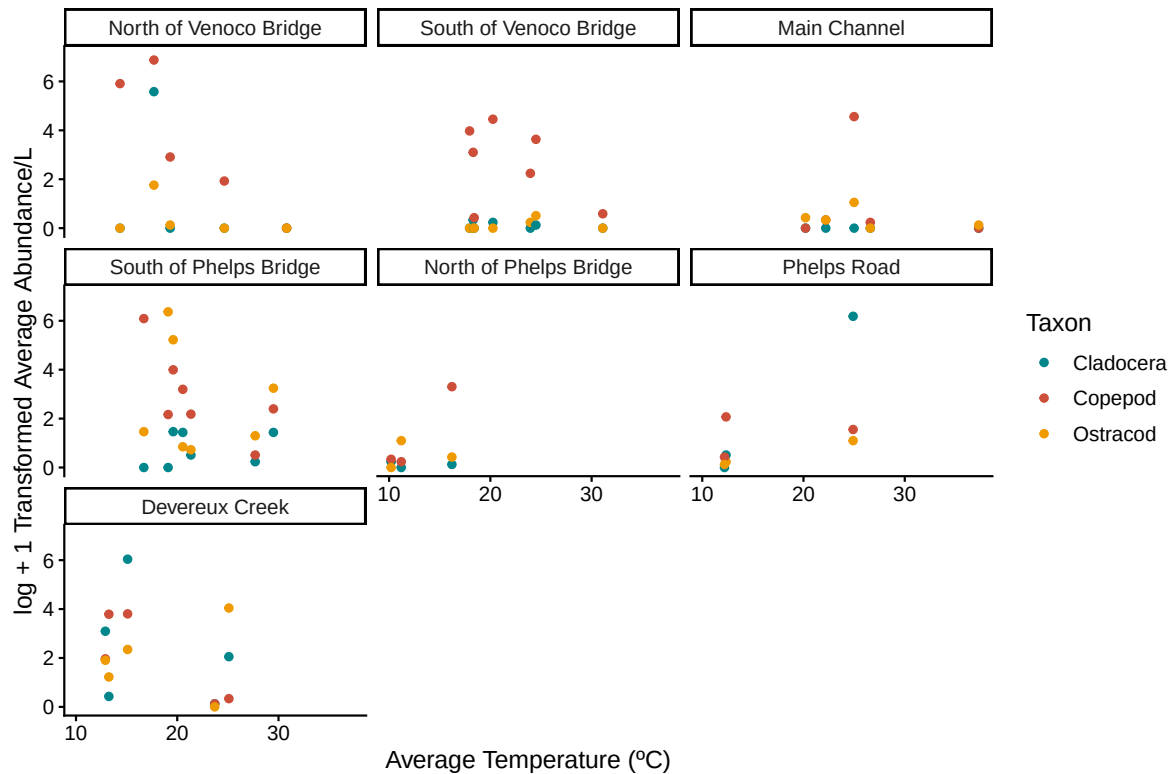


Figure 2

```
# base layer: create new object to store figure 2
figure2 <- ggplot(data = aq_ins_clean,
  aes(x = taxon,
    y = log_ave_lit,
    color = taxon)) +

# first layer: jitter plot
geom_jitter(height = 0,
  alpha = 1,
  width = 0.2,
  shape = 21) +

# second layer: points to represent medians
```

```

stat_summary(geom = "point",
             fun = "mean",
             size = 4,
             color = "black",
             shape = 18) + # 18 = filled diamond

stat_summary(geom = "errorbar",
             fun.data = mean_se,
             width = 0.1,
             color = "black") +

# colors
scale_color_manual(values = taxon_cols) +

# titles
labs(x = "Taxon",
     y = "log + 1 transformed average abundance/L",
     title = "Figure 2:

Mean and Distribution of Average Abundance for Three Dominant Taxa")

figure2

```



Figure 2:

Mean and Distribution of Average Abundance for Three Dominant Taxa

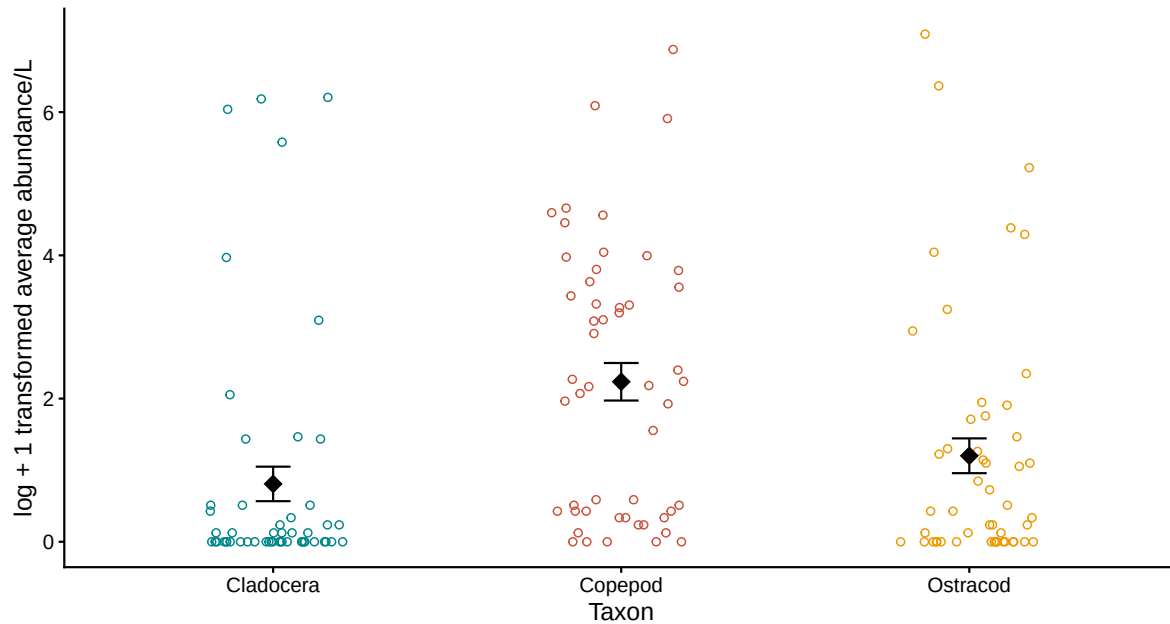


Figure 3a

```
aq_ins_clean <- aq_ins_clean |>
  mutate(temp_range = cut(med_temp,
    breaks = c(10.0, 15.6, 21.2, 26.8, 32.4, 38.0),
    labels = c("10.0-15.6C", "15.6-21.2C",
      ↪ "21.2-26.8C",
      ↪ "26.8-32.4C", "32.4-38.0C"),
    include.lowest = TRUE),
  point_color = if_else(log_ave_lit == 0, "Zero Abundance",
    ↪ as.character(temp_range)),
  point_color = factor(point_color, levels = c("10.0-15.6C",
    ↪ "15.6-21.2C",
    ↪ "21.2-26.8C",
    ↪ "26.8-32.4C",
    ↪ "32.4-38.0C", "Zero Abundance"))))
```

```

# base layer: create new object to store figure 3a
figure3a <- ggplot(data = aq_ins_clean,
                   mapping = aes(x = med_temp,
                                y = log_ave_lit,
                                color = temp_range)) +

# first layer: points
geom_point() +

# facet by taxon
facet_wrap(~taxon,
           scales = "free") +

# add legend
theme(legend.position = "right")+

# labels
labs(title = "Figure 3a:
Average Abundance and Average Water Temperature for Three Dominant Taxa",
      x = "Average Temperature (°C)",
      y = "log + 1 Transformed Average Abundance/L",
      color = "Temperature Range") +

scale_x_continuous(
  breaks = c(10.0, 15.6, 21.2, 26.8, 32.4, 38.0),
  limits = c(10.0, 38.0)) +

scale_color_manual(values = c(
  "10.0-15.6C" = "steelblue",
  "15.6-21.2C" = "skyblue",
  "21.2-26.8C" = "gold",
  "26.8-32.4C" = "orange",
  "32.4-38.0C" = "tomato3",
  "Zero Abundance" = "black"
), na.value = "grey50")

figure3a

```

Figure 3a:

Average Abundance and Average Water Temperature for Three Dominant Taxa

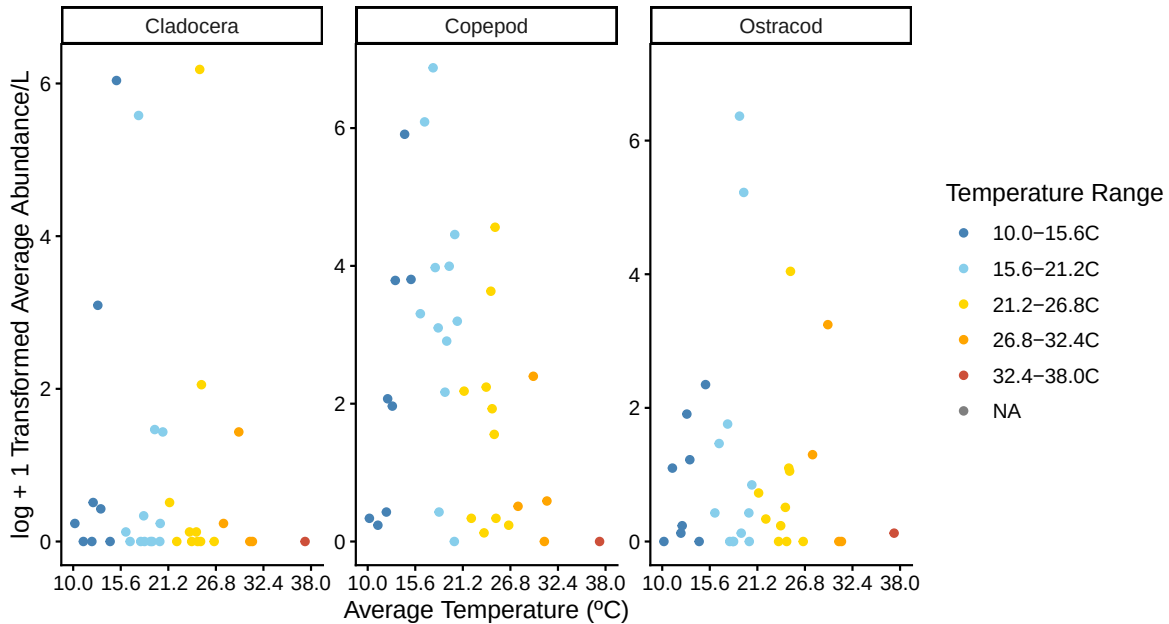


Figure 3b

```
# base layer: create new object to store figure 3b
figure3b <- ggplot(data = aq_ins_clean,
  mapping = aes(x = site,
    y = log_ave_lit,
    color = site)) +

# first layer: points
geom_point() +

# facet by taxon
facet_wrap(~taxon) +

# add legend
theme(legend.position = "right") +

# manual colors
scale_color_manual(values = sites_cols) +
```

```
# labels and title
labs(title = "Figure 3b:

Average Abundance and Average Water Temperature for Three Dominant Taxa by
↪ Site",
      x = "Average Temperature (°C)",
      y = "log + 1 Transformed Average Abundance/L",
      color = "Site")
```

figure3b

Figure 3b:

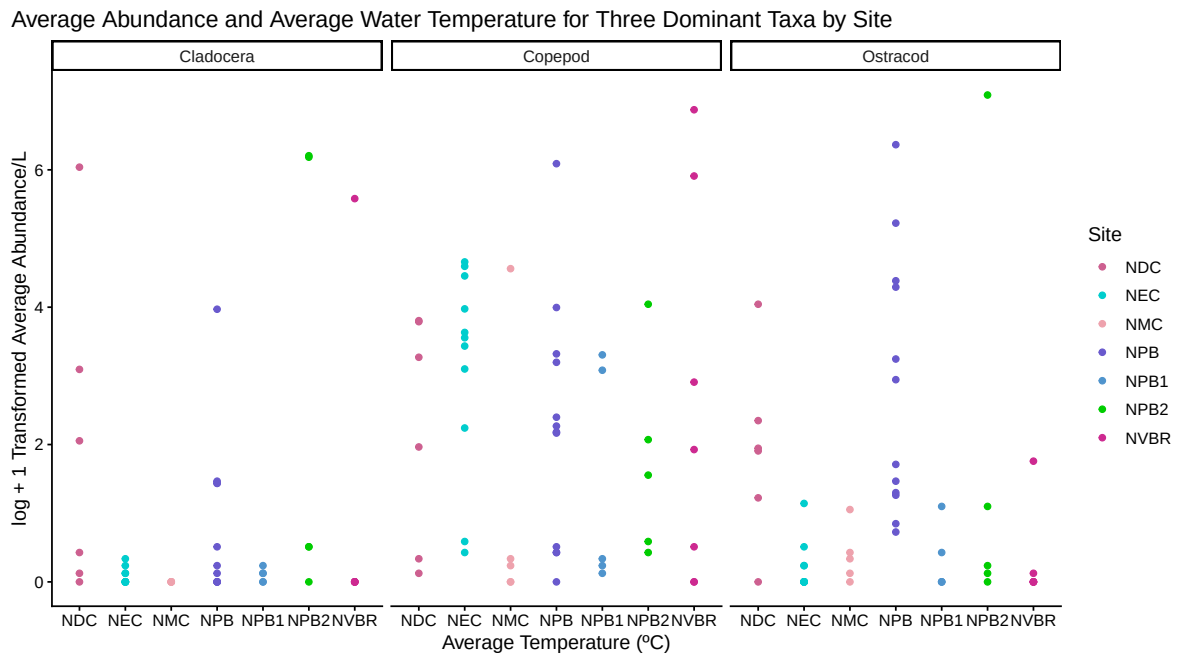


Figure 4

```
# base layer: create new object to store figure 4
figure4 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
                                y = log_ave_lit,
```

```

                                color = taxon)) +
# first layer: points
geom_point() +

# facet by season
facet_wrap(~season,
           scales = "free") +

# titles and labels
labs(title = "Figure 4:
Average Abundance and Average Water Temperature by Season for Three Dominant
↪ Taxa",
      x = "Average Temperature (°C)",
      y = "log + 1 Transformed Average Abundance/L",
      color = "Taxon") +

# manual colors
scale_color_manual(values = taxon_cols) +

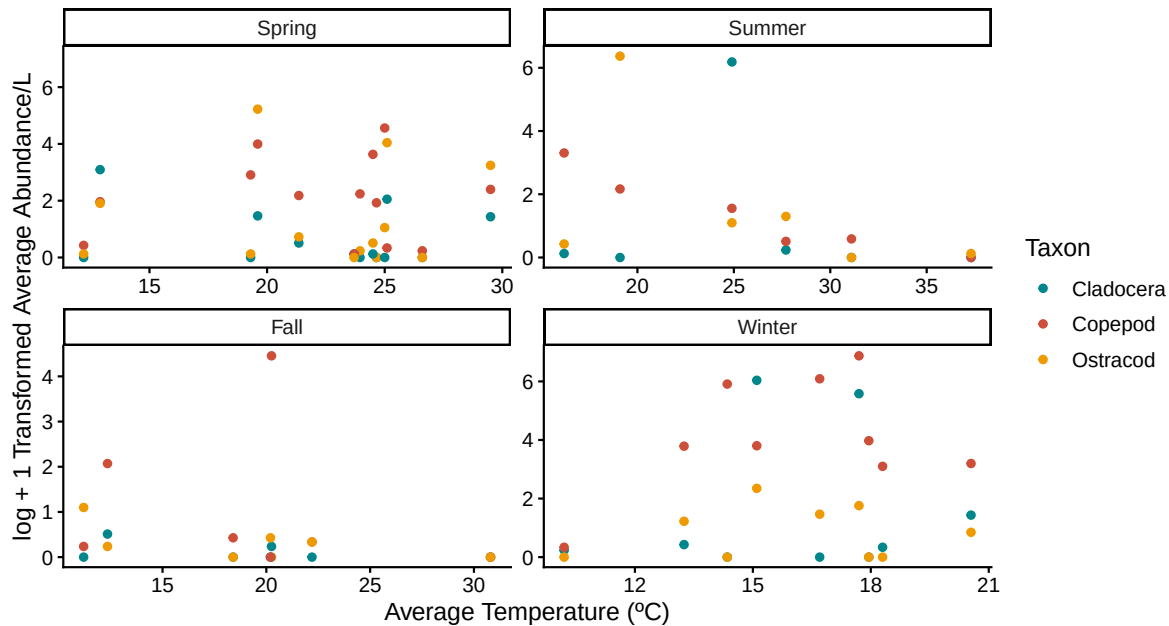
# legend
theme(legend.position = "right")

figure4

```

Figure 4:

Average Abundance and Average Water Temperature by Season for Three Dominant Taxa



## Other exploratory visualizations

```
# creating new object for ridgeline
ridges <- ggplot(data = aq_ins_clean,
  # x-axis: avg temp
  # y-axis: avg abund
  # fill geometries by site
  mapping = aes(x = log_ave_lit,
    y = taxon,
    fill = taxon)) +
  # ridges (scale makes ridges shorter so that they don't overlap)
  geom_density_ridges(scale = 0.8) +

  # adding means and 95% confidence intervals
  stat_summary(geom = "pointrange",
    fun.data = mean_cl_normal,
    size = 0.75,
    linewidth = 1) +
  # filling geometries with color
```

```

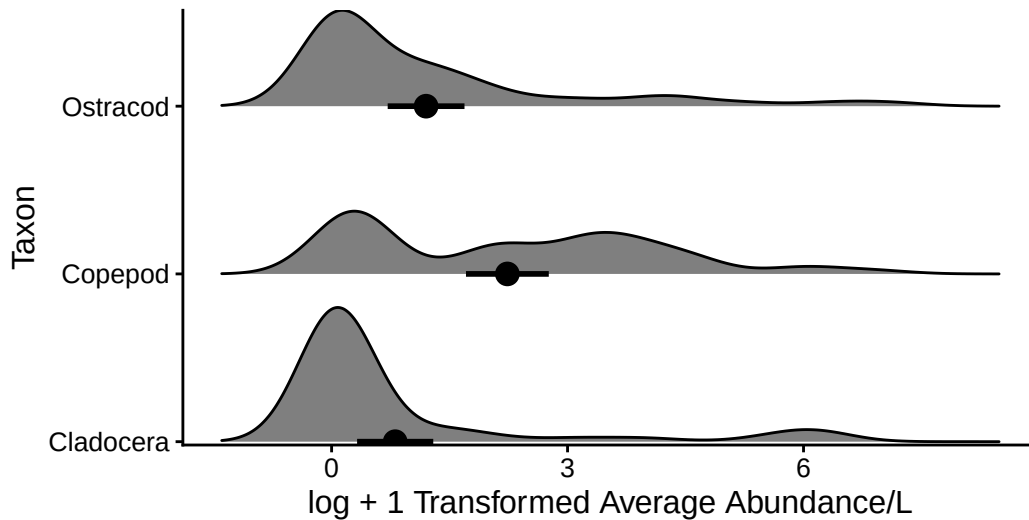
scale_fill_manual(values = sites_cols) +
# setting y-axis expansions
scale_y_discrete(expand = expansion(mult = c(0.01, 0.15))) +
# relabelling x- and y-axis
labs(title = "Exploratory Figure 5:
Ridge Figure Average Abundance for Three Dominant Taxa",
      y = "Taxon",
      x = "log + 1 Transformed Average Abundance/L")

# displaying object
ridges

```

Exploratory Figure 5:

### Ridge Figure Average Abundance for Three Dominant Taxa



### Ideas of Analysis for Background

Background about NCOS and project history:

From 1966 to 2013, the space where NCOS sits was used as a 9-hole golf course. After a couple years of extensive planning, the NCOS restoration project began in February 2017, managed by the UCSB Cheadle Center. This space has created more than 40 acres of estuarine and palustrine wetlands and restored more than 60 acres of upland habitats. In comparison with

COPR, the study has found that NCOS appears to have equivalent, if not slightly greater species richness and evenness.

Specifics about aquatic invertebrate:

Samples are collected by student interns and volunteers using a filtered bucket and dip net method, with the 3 most common invertebrates found with the filtered beaker 250 micron method being copepods, cladocera, ostracods and the 3 most common invertebrates found with the dipnet 500 micron method being copepods, ostracods, hemipteras. There is a strong negative linear relationship between aquatic invertebrate and salinity, as shown by a 2022 monitoring report of Devereux slough. Additionally found in this report, “from the data so far, the Slough and its associated ponds and vernal pools contain a fairly small set of invertebrate inhabitants. While we will need more time to determine what NCOS’s steady-state environment will be like, COPR’s portion of the Slough has some relatively extreme conditions with regard to Salinity, Temperature, and Dissolved Oxygen”. Only four planktonic taxa appear in any great abundance: Copepod, Corixidae, Ostracod, and Cladocera. The benthic substrate has a much higher concentration of these taxa, which is where the dip net testing is now most focusing on. In the 2024 Aquatic Invertebrate Assessment at the North Campus Open Space, it was noted that the highest species diversity was found at the Phelp’s creek site compared to a couple other NCOS sites. There are other reports worth exploring that compare the linkage between aquatic invertebrate and other animals, such as waterbirds.

Important Questions and relationships that can be explored:

Why is this data important; why are these species important? What do they indicate about the health of the area, and what is their role among the trophic levels? What is the relationship between aquatic invertebrates and temperature, and what does this mean about the health of the ecosystem? Does abundance vary across site or across season? Is there a relationship between salinity and temperature? The coding that explores these relationships is listed above during the creation of the visualizations.

Links to reports:

- <https://escholarship.org/uc/item/16p046mb#main>
- <https://escholarship.org/uc/item/59c872mm>
- <https://escholarship.org/uc/item/64f0w6hx>
- <https://escholarship.org/uc/item/6n78q34g>
- <https://escholarship.org/uc/item/1bc3t7fb>

## **Plan for elective**

### **Timeline**

Week 7:



- Research all three organisms (copepod, ostracod, cladoceran)
- Gather reference images
- Review your figures
- Take notes on what each figure shows about the species
- Brainstorm collage/drawing style and decide on materials (digital vs. physical).

Week 8 (timeline check-in):

- Have a rough sketch/draft of all three 2D collages– one per organism
- Bring your notes on what the figures show about each species. This is our plan and draft milestone.

Week 9:

- Refine and finalize the artwork for 1–2 of the organisms
- Fill in your species notes with complete observations from your figures.

Week 10:

- Complete the third organism's artwork and finish all species notes
- Do a full review, make sure each collage clearly reflects what your figures show about that species.

Finals week:

- Final polish, compile everything together
- Present your completed 2D collage triptych with notes.

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## Visualization Approach and Temperature Color Scale

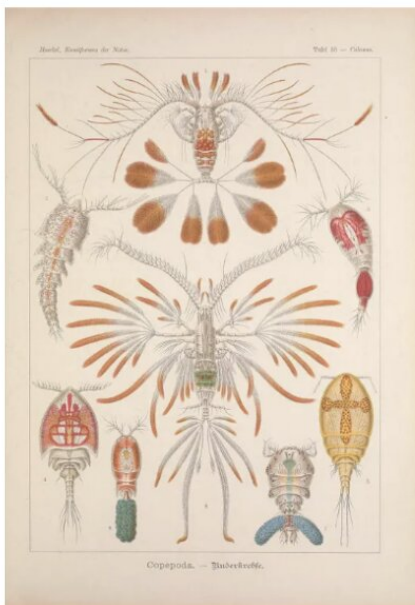
For our final visualization, we decided on a physical drawing. Each organism will be represented by an outline, filled with paint and pastels to illustrate the relationship between abundance and temperature. The colors inside each outline correspond to a temperature-based heat range, divided into five equal intervals spanning the full range of recorded temperatures (10.20°C – 37.30°C):

- 10.0°C – 15.6°C
- 15.6°C – 21.2°C
- 21.2°C – 26.8°C
- 26.8°C – 32.4°C
- 32.4°C – 38.0°C

## Gathering Images and Research on Organisms

### Copepods

- Group of small crustaceans
- Found in nearly every salt and freshwater habitat
- Some are planktonic, some benthic (live on sediments)
- Teardrop shaped body with large antennae
- Thin exoskeleton is transparent; they eat phytoplankton, detritus, and small zooplankton
- Several species are bioluminescent
- They jump out of the water to evade predators



#### Ostracods

- tiny, ubiquitous crustaceans ranging from 0.1 to 30 mm in size
- Commonly known as “seed shrimp”
- Bivalve shells that encase their bodies
- About 33,000 species, some being bioluminescent
- Most common arthropods in the fossil record



## Cladocera

- Commonly known as the water flea
- Freshwater crustaceans, feeding on microscopic organics, though some forms are predatory
- Over 1000 species
- Down-turned head with a single median compound eye, and a carapace covering the apparently unsegmented thorax and abdomen
- Most cladoceran species live in fresh water and other inland water bodies

